# STRANGE QUANTUM ATOMS

Most atoms are in a state of constant motion, which makes their properties hard to measure precisely. To get the most accurate measurements, physicists cool them to extremely low temperatures, where the random movement of particles due to excess kinetic (motion) energy is minimized. This can be done by using laser beams to slow down the atoms, ultimately reaching temperatures close to absolute zero (–459.67°F/–273.15°C), where strange quantum properties can develop.

**COOLING LASERS**

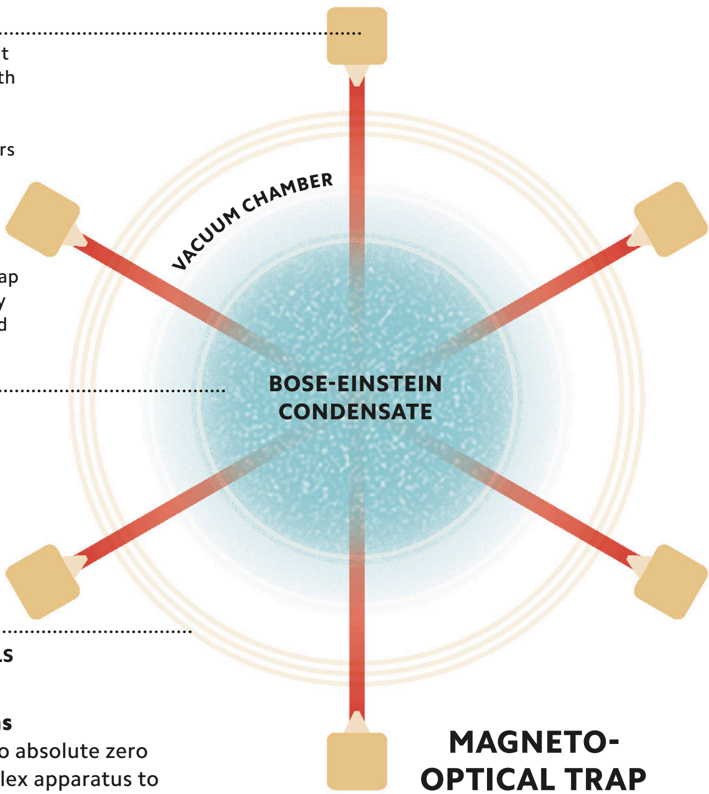
Lasers are "tuned" so that they will interact only with atoms moving toward them. Each collision with a photon from these lasers drains more momentum from the atom.

Atoms are fed into the trap at low pressure, gradually accumulating into a cloud whose properties can be studied (see pp.78–79).

**COOLED ATOM CLOUD**

A varying magnetic field is designed to channel the atoms in directions that ensure the cooling lasers have maximum effect.

**ANTI-HELMHOLTZ COILS**




## MAGNETO-OPTICAL TRAP

**Slowing down atoms**
Cooling atoms close to absolute zero involves using a complex apparatus to bring their motion to a near-complete stop in a magneto-optical trap.